

Practice Questions for Convex Combination

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1 Questions

1. A convex combination of points $x_1, x_2, \dots, x_n$ in R^d is a point x of the form $x = \sum_{i=1}^n \lambda_i x_i$, where the coefficients λ_i satisfy certain conditions. What are these conditions?
 - a. $\sum_{i=1}^n \lambda_i = 1$
 - b. $\lambda_i \geq 0$ for all $i = 1, \dots, n$
 - c. $\sum_{i=1}^n \lambda_i = 1$ and $\lambda_i \geq 0$ for all $i = 1, \dots, n$
 - d. $\sum_{i=1}^n \lambda_i \leq 1$ and $\lambda_i \geq 0$ for all $i = 1, \dots, n$
 - e. $\sum_{i=1}^n \lambda_i = 1$ and $\lambda_i \leq 1$ for all $i = 1, \dots, n$
2. Why is the concept of a convex combination important in the study of convex sets?
 - a. Convex combinations always result in points outside the convex set.
 - b. Convex combinations are irrelevant to the study of convex sets.
 - c. Convex combinations always result in points on the boundary of the convex set.
 - d. Convex combinations of points within a convex set always remain within the convex set.
 - e. Convex combinations are only defined for non-convex sets.
3. What is the geometric interpretation of a convex combination of two points in R^2 ?
 - a. A point outside the line segment connecting the two points.
 - b. A point on the line passing through the two points, but outside the line segment connecting them.
 - c. Any point on the line passing through the two points.
 - d. A point on the line segment connecting the two points.
 - e. A point that is equidistant from the two points.
4. Suppose you have a set of points S in R^d . The convex hull of S , denoted as $\text{conv}(S)$, is the smallest convex set containing all points in S . How can you describe $\text{conv}(S)$ in terms of convex combinations?
 - a. $\text{conv}(S)$ is the set of all points that are not convex combinations of points in S .
 - b. $\text{conv}(S)$ is the set of all points that are linear combinations of points in S .
 - c. $\text{conv}(S)$ is the set of all points that are convex combinations of points in S .

- d. $\text{conv}(S)$ is the set of all points that are affine combinations of points in S .
 - e. $\text{conv}(S)$ is empty if S is not a convex set.
5. Explain the concept of a convex combination and its significance in the context of convex sets.
- a. A convex combination is a linear combination of points where the coefficients are all non-negative and sum to one. It's significant because any convex combination of points within a convex set remains within the set.
 - b. A convex combination is a weighted average of points where the weights are all positive. It's significant because it helps define the boundary of a convex set.
 - c. A convex combination is a linear combination of points where the coefficients sum to one. It's significant because it allows us to express any point in a convex set as a combination of its extreme points.
 - d. A convex combination is a weighted average of points where the weights are non-negative. It's significant because it's used to prove that the feasible region in linear programming is convex.
 - e. A convex combination is a linear combination of points where the coefficients are all positive and sum to one. It's significant because it defines the interior of a convex set.
6. Given the points $x_1 = (1, 2)$ and $x_2 = (3, 4)$ in R^2 , find a convex combination of x_1 and x_2 such that the resulting point lies on the line segment connecting x_1 and x_2 .
- a. $(2, 3)$
 - b. $(0, 0)$
 - c. $(4, 6)$
 - d. $(1, 1)$
 - e. $(5, 6)$
7. What is a convex combination, and how does it relate to the concept of a convex set?
- a. A convex combination is a linear combination of points where the coefficients are all positive and sum to one. It relates to convex sets because any convex combination of points within a convex set must also lie within the set.
 - b. A convex combination is a weighted average of points where the weights are non-negative and sum to one. It relates to convex sets because a set is convex if and only if it contains all convex combinations of its points.
 - c. A convex combination is a linear combination of points where the coefficients are non-negative and sum to one. It is unrelated to convex sets.
 - d. A convex combination is a weighted average of points where the weights are positive and sum to one. A set is convex if it contains at least one convex combination of its points.
 - e. A convex combination is any linear combination of points. It relates to convex sets because the line segment connecting any two points in a convex set is also in the set.
8. Explain Carathéodory's Theorem and its significance in the context of linear programming.

- a. Carathéodory's Theorem states that any point in the convex hull of a set in R^m can be expressed as a convex combination of at most m points from the set. This is significant because it limits the number of points needed to represent any point in the convex hull, improving computational efficiency in linear programming.
 - b. Carathéodory's Theorem states that any point in the convex hull of a set in R^m can be expressed as a convex combination of at most $m + 1$ points from the set. This is significant because it limits the number of points needed to represent any point in the convex hull, improving computational efficiency in linear programming.
 - c. Carathéodory's Theorem states that any point in a convex set can be expressed as a convex combination of its extreme points. This is significant because it simplifies the representation of convex sets in linear programming.
 - d. Carathéodory's Theorem is unrelated to linear programming.
 - e. Carathéodory's Theorem states that any point in the convex hull of a set can be expressed as a convex combination of all points in the set. This is significant because it provides a complete characterization of the convex hull.
9. Given two points $z_1 = (1, 2)$ and $z_2 = (3, 4)$ in R^2 , what is a convex combination of z_1 and z_2 where the resulting point has an x-coordinate of 2?
- a. $\frac{1}{2}z_1 + \frac{1}{2}z_2$
 - b. $\frac{1}{3}z_1 + \frac{2}{3}z_2$
 - c. $\frac{2}{3}z_1 + \frac{1}{3}z_2$
 - d. $\frac{3}{4}z_1 + \frac{1}{4}z_2$
 - e. $\frac{1}{4}z_1 + \frac{3}{4}z_2$
10. What are the conditions on the coefficients λ_i for a point $x = \sum_{i=1}^n \lambda_i x_i$ to be a convex combination of points $x_1, x_2, \dots, x_n$ in R^d ?
- a. $\lambda_i \geq 0$ for all i
 - b. $\sum_{i=1}^n \lambda_i = 1$
 - c. $\lambda_i \geq 0$ for all i and $\sum_{i=1}^n \lambda_i = 1$
 - d. $\sum_{i=1}^n \lambda_i \leq 1$
 - e. $\lambda_i \leq 1$ for all i

2 Answers

1. A convex combination of points $x_1, x_2, \dots, x_n$ in R^d is a point x of the form $x = \sum_{i=1}^n \lambda_i x_i$, where the coefficients λ_i satisfy certain conditions. What are these conditions?
 - a. This condition only ensures that the weights sum to 1, but it doesn't guarantee that they are non-negative, which is also a requirement for a convex combination.
 - b. This condition only ensures that the weights are non-negative, but it doesn't guarantee that they sum to 1, which is also a requirement for a convex combination.
 - c. A convex combination requires both conditions: the weights must sum to 1, and they must be non-negative. This ensures the combined point lies within the convex hull of the original points.
 - d. This allows for weights that sum to less than 1, which is not a characteristic of a convex combination. The weights must sum to exactly 1.
 - e. This condition allows for negative weights, which is not allowed in a convex combination. The weights must be non-negative.

2. Why is the concept of a convex combination important in the study of convex sets?
 - a. This is incorrect; convex combinations of points within a convex set always remain within the set.
 - b. This is incorrect; convex combinations are fundamental to the definition and properties of convex sets.
 - c. This is incorrect; convex combinations can result in points anywhere within the convex set, not just on the boundary.
 - d. The definition of a convex set is that any convex combination of points within the set also lies within the set. This is a fundamental property used to characterize and analyze convex sets.
 - e. This is incorrect; convex combinations are specifically defined for convex sets.

3. What is the geometric interpretation of a convex combination of two points in R^2 ?
 - a. This is incorrect; a convex combination always lies on the line segment connecting the points.
 - b. This is incorrect; a convex combination always lies on the line segment connecting the points.
 - c. This is incorrect; a convex combination is restricted to the line segment between the points.
 - d. A convex combination of two points in R^2 represents a point on the line segment connecting those two points. The weights in the combination determine the position of the point along the segment.
 - e. This is incorrect; the position of the point is determined by the weights in the convex combination, not by equidistance.

4. Suppose you have a set of points S in R^d . The convex hull of S , denoted as $\text{conv}(S)$, is the smallest convex set containing all points in S . How can you describe $\text{conv}(S)$ in terms of convex combinations?
 - a. This is incorrect; the convex hull is the set of all points that *are* convex combinations of points in S .
 - b. This is incorrect; linear combinations can include negative weights, which are not allowed in convex combinations.

- c. The convex hull of a set S is precisely the set of all possible convex combinations of points from S . This captures the smallest convex set containing all points in S .
 - d. This is incorrect; affine combinations allow for weights that sum to 1 but can include negative weights, unlike convex combinations.
 - e. This is incorrect; the convex hull is always defined, even if S is not convex. It is the smallest convex set containing S .
5. Explain the concept of a convex combination and its significance in the context of convex sets.
- a. A convex combination is indeed a linear combination of points where the coefficients are non-negative and sum to one. This is the precise definition. Its significance lies in the fact that if a set is convex, any convex combination of points within that set will also be within the set. This property is fundamental to the study of convex sets and is crucial in linear programming, where the feasible region is always a convex set.
 - b. While a convex combination is a weighted average, the weights must be non-negative, not necessarily positive. Also, it doesn't define the boundary alone; the boundary can be more complex than just convex combinations.
 - c. The coefficients summing to one is correct, but the coefficients must also be non-negative. While expressing points in a convex set as combinations of extreme points is related, it's not the primary significance of the definition itself.
 - d. The weights being non-negative is correct, but the fact that it's used to prove the convexity of the feasible region is a consequence of the definition, not the definition's primary significance.
 - e. The coefficients must be non-negative, not necessarily positive. Also, convex combinations don't define only the interior; they can include boundary points as well.
6. Given the points $x_1 = (1, 2)$ and $x_2 = (3, 4)$ in $\mathbb{R}^2$, find a convex combination of x_1 and x_2 such that the resulting point lies on the line segment connecting x_1 and x_2 .
- a. A convex combination is of the form $\lambda x_1 + (1 - \lambda)x_2$ where $0 \leq \lambda \leq 1$. If we let $\lambda = 0.5$, we get $0.5(1, 2) + 0.5(3, 4) = (2, 3)$. This point lies on the line segment connecting $(1, 2)$ and $(3, 4)$.
 - b. $(0, 0)$ is not a convex combination of $(1, 2)$ and $(3, 4)$ because it does not lie on the line segment connecting them. A convex combination must be a weighted average of the points, with weights between 0 and 1.
 - c. $(4, 6)$ is not a convex combination of $(1, 2)$ and $(3, 4)$ because it lies outside the line segment connecting them. Extrapolating the line segment would require λ to be greater than 1, which is not allowed in a convex combination.
 - d. $(1, 1)$ is not a convex combination of $(1, 2)$ and $(3, 4)$ because it does not lie on the line segment connecting them. A convex combination must be a weighted average of the points, with weights between 0 and 1.
 - e. $(5, 6)$ is not a convex combination of $(1, 2)$ and $(3, 4)$ because it lies outside the line segment connecting them. Extrapolating the line segment would require λ to be greater than 1, which is not allowed in a convex combination.
7. What is a convex combination, and how does it relate to the concept of a convex set?
- a. While the coefficients must sum to one, they can be zero. A convex combination of points within a convex set must lie within the set, but this is a consequence of the definition, not the definition itself.

- b. A convex combination is a weighted average of points where the weights are non-negative and sum to one. This directly relates to the definition of a convex set: a set is convex if and only if it contains all convex combinations of its points. If a set contains all convex combinations of its points, then the line segment connecting any two points in the set is also in the set, satisfying the definition of convexity.
 - c. A convex combination is directly related to the definition of a convex set. A set is convex if and only if it contains all convex combinations of its points.
 - d. The weights must be non-negative, not necessarily positive. Also, a set is convex if it contains *all* convex combinations of its points, not just at least one.
 - e. A convex combination requires non-negative coefficients that sum to one. While the line segment property is a consequence of convexity, it's not the definition of a convex combination.
8. Explain Carathéodory's Theorem and its significance in the context of linear programming.
- a. The theorem states that at most $m + 1$ points are needed, not m .
 - b. Carathéodory's Theorem states that any point in the convex hull of a set in R^m can be expressed as a convex combination of at most $m + 1$ points from the set. This is significant in linear programming because the feasible region is often a convex set, and the theorem implies that we don't need to consider all possible convex combinations of points in the feasible region to find an optimal solution; we only need to consider combinations of at most $m + 1$ points. This significantly reduces the computational burden.
 - c. While related to convex sets, this statement is not Carathéodory's Theorem. Carathéodory's Theorem applies to any point in the convex hull, not just extreme points.
 - d. Carathéodory's Theorem is a fundamental result in convex geometry with direct implications for linear programming algorithms.
 - e. The theorem states that at most $m + 1$ points are needed, not all points in the set.
9. Given two points $z_1 = (1, 2)$ and $z_2 = (3, 4)$ in R^2 , what is a convex combination of z_1 and z_2 where the resulting point has an x-coordinate of 2?
- a. This is incorrect. While this is a valid convex combination, the x-coordinate is 2, not $5/3$.
 - b. This is incorrect. The x-coordinate of this convex combination is $\frac{7}{3}$, not 2.
 - c. Let the convex combination be $tz_1 + (1-t)z_2$. The x-coordinate is $t(1) + (1-t)(3) = 2$. Solving for t , we get $t + 3 - 3t = 2$, which simplifies to $-2t = -1$, so $t = \frac{1}{2}$. Therefore, the convex combination is $\frac{1}{2}z_1 + \frac{1}{2}z_2$. However, this is not among the options. Let's check option C: $\frac{2}{3}(1, 2) + \frac{1}{3}(3, 4) = (\frac{2}{3} + 1, \frac{4}{3} + \frac{4}{3}) = (\frac{5}{3}, \frac{8}{3})$. The x-coordinate is $\frac{5}{3} \neq 2$. Let's try option B: $\frac{1}{3}(1, 2) + \frac{2}{3}(3, 4) = (\frac{1}{3} + 2, \frac{2}{3} + \frac{8}{3}) = (\frac{7}{3}, \frac{10}{3})$. The x-coordinate is $\frac{7}{3} \neq 2$. Let's check option A again: $t(1) + (1-t)(3) = 2$, which gives $t = 1/2$. Then the convex combination is $\frac{1}{2}(1, 2) + \frac{1}{2}(3, 4) = (2, 3)$. The x-coordinate is 2. This is a convex combination because $t = 1/2$ and $1 - t = 1/2$, and both are between 0 and 1 and sum to 1. Therefore, option A is correct.
 - d. This is incorrect. The x-coordinate of this convex combination is $\frac{7}{4}$, not 2.
 - e. This is incorrect. The x-coordinate of this convex combination is $\frac{11}{4}$, not 2.
10. What are the conditions on the coefficients λ_i for a point $x = \sum_{i=1}^n \lambda_i x_i$ to be a convex combination of points $x_1, x_2, \dots, x_n$ in R^d ?

- a. This is incorrect because it only specifies the non-negativity condition; the coefficients must also sum to 1.
- b. This is incorrect because it only specifies the summation condition; the coefficients must also be non-negative.
- c. A convex combination requires that all coefficients are non-negative and sum to 1. This ensures that the resulting point lies within the convex hull of the given points.
- d. This is incorrect. The coefficients must sum to exactly 1, not less than or equal to 1.
- e. This is incorrect because it only specifies an upper bound on the coefficients; they must also be non-negative and sum to 1.